

# Mathematics for Social Scientists

## Part 4: Introduction to Probability Theory

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# Outline

Counting and Combinatorics

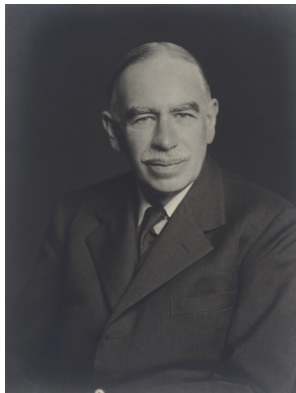
Probabilistic Formalism

Conditional Probability and Bayes Rule

Independence

Mathematical probability is a systematic, rigorous treatment of uncertainty.

It was developed in the 17<sup>th</sup> and 18<sup>th</sup> centuries, due to an interest in gambling by various nobles and the mathematicians they employed. Most modern formalism was developed by Keynes and Kolmogorov in the early 20<sup>th</sup> century.



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## Counting: a motivating example

### Example.

3 people  $a$ ,  $b$  and  $c$  live together.

Every week 2 people should clean the kitchen and bathroom. What are the choices?

- One person cleans the bathroom, the other the kitchen.

The set of choices for  $\{\text{bathroom, kitchen}\}$  is

$$\{\{a, b\}, \{b, a\}, \{a, c\}, \{c, a\}, \{b, c\}, \{c, b\}\}.$$

- Both people cooperate to clean both rooms.

The set of choices is

$$\{\{a, b\}, \{a, c\}, \{b, c\}\}.$$

The question “what is the probability that  $a$  is cleaning the kitchen this week?” has a different answer in the different regimes.

# Why counting?

- Many probability questions are really *counting* questions.
- In a finite sample space with equally likely outcomes:

$$\mathbb{P}(A) = \frac{\#\{\text{favourable outcomes}\}}{\#\{\text{all outcomes}\}}.$$

- So we need systematic tools to count outcomes:
  - how many possible **orders**?
  - how many possible **subsets**?
- This is the role of **permutations** and **combinations**.

## Factorials and “ $n$ choose $k$ ”

- For a natural number  $n \in \mathbb{N}$ , the **factorial** of  $n$  is

$$n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1, \quad 0! := 1.$$

- The **binomial coefficient** “ $n$  choose  $k$ ” counts how many  $k$ -element subsets we can select from  $n$  elements:

$$\binom{n}{k} = \binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad 0 \leq k \leq n.$$

- Example: how many ways can a voter select 3 candidates from a field of 7?

$$\binom{7}{3} = \frac{7!}{3!4!} = \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} = 35.$$

# Counting without panic: two key questions

In many exercises, you can decide the right formula by answering two questions:

## 1. Does order matter?

- e.g. list of speakers, queue at the ticket counter  $\Rightarrow$  order matters.
- e.g. committee of 3 out of 7 people  $\Rightarrow$  order does not matter.

## 2. Do we sample with or without replacement?

- with replacement: we can pick the same element multiple times (e.g. draw with replacement from an urn).
- without replacement: once an element is taken, it is not available any more (e.g. 5 cards from a deck, survey sample without replacement).



# Permutations and combinations

|                                               | with replacement                                                           | without replacement                                                            |
|-----------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Permutation</b><br>(order matters)         | $n^k$<br>$k$ -tuples from $n$ objects                                      | $\binom{n}{k} k! = \frac{n!}{(n-k)!}$<br>order $k$ out of $n$ distinct objects |
| <b>Combination</b><br>(order does not matter) | $\binom{n+k-1}{k} = \frac{(n+k-1)!}{k!(n-1)!}$<br>$k$ draws from $n$ types | $\binom{n}{k} = \frac{n!}{k!(n-k)!}$<br>choose $k$ out of $n$ objects          |

Here  $n$  is the number of available objects (population size) and  $k$  the number drawn.

## Example: queue at the train station

### Permutations without replacement

There are 4 passengers waiting to buy tickets. In how many different sequences can the tickets be sold?

- We have  $n = 4$  people and want to order all of them:  $k = n = 4$ .
- Order matters (A-B-C-D is different from B-A-C-D), and we do not reuse people.
- So the number of possible queues is

$$\frac{4!}{(4-4)!} = 4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24.$$

## Example: 5-card hands from a deck

### Combinations without replacement

A standard deck has 52 cards. How many different 5-card hands are possible?

- We choose  $k = 5$  cards out of  $n = 52$  cards.
- Order does not matter (a “hand” is just a set of cards).
- We sample without replacement (we do not put cards back into the deck).
- So the number of possible 5-card hands is

$$\binom{52}{5} = \frac{52!}{5!47!}.$$

In a finite sample space with equally likely 5-card hands, a probability is

$$\mathbb{P}(A) = \frac{\#\{\text{hands with property } A\}}{\binom{52}{5}}.$$

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**Probabilistic Formalism**

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# Experiments, sample spaces, events

## Definition 1.

- An **experiment** is a process that produces an outcome we cannot predict with certainty in advance.
- The **sample space**  $S$  (also written  $\Omega$ ) is the set of all possible outcomes of the experiment.
- An **event** is a subset  $A \subseteq S$ , i.e. a collection of outcomes.

## Example (Coins and dice).

- Flip a single coin:  $S = \{H, T\}$ , events are subsets of  $S$ , e.g.  $\{H\}$ ,  $\{T\}$ ,  $S$ .
- Roll a die:  $S = \{1, 2, 3, 4, 5, 6\}$ , events are subsets of  $S$ , e.g.  $\{1\}$ ,  $\{2, 4, 6\}$ ,  $S$ .

## Probability: axioms

### Definition 2 (Probability).

A **probability** on a sample space  $S$  with event set  $E$  is a function

$$P : E \rightarrow [0, 1]$$

that satisfies the following three axioms (Kolmogorov's axioms):

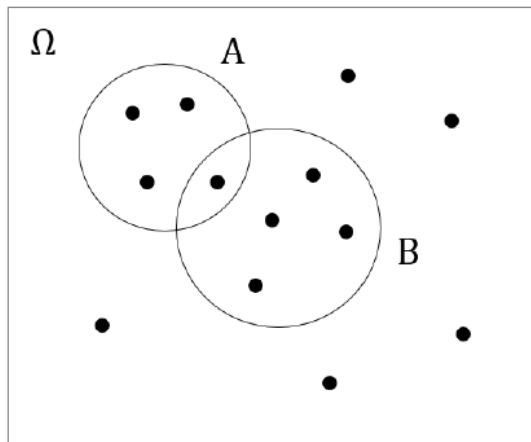
- (1) For any event  $A$ ,  $P(A) \geq 0$ .
- (2)  $P(S) = 1$ .
- (3) For any sequence of disjoint events  $A_1, A_2, \dots$ ,

$$P\left(\bigcup_{i=1}^k A_i\right) = \sum_{i=1}^k P(A_i) \quad (\text{finite case}).$$

- Probability is a function that maps events to a real number.
- Probabilities are non-negative, add up to 1 over the whole sample space, and add over disjoint events.

# Probability as a measure

- The sample space  $S$  (or  $\Omega$ ) is the whole rectangle (total probability 1).
- Events  $A$  and  $B$  are subsets of  $S$  (circles).
- The points are individual outcomes  $\omega \in S$ .
- The probability measure  $P(\cdot)$  assigns a “size” to each event, consistent with the axioms.



# Basic rules of probability

## Theorem 1 (Basic rules).

Let  $A, B \subseteq S$  be events. Then:

- $P(\emptyset) = 0$ .
- $0 \leq P(A) \leq 1$ .
- **Complement:**  $P(A^c) = 1 - P(A)$ .
- **Monotonicity:** If  $A \subseteq B$  then  $P(A) \leq P(B)$ .
- **Union of two events:**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

These follow from the axioms and give us the standard “probability operations” used later.



## Classical probability on a finite sample space

In many examples, the sample space  $S$  is finite and all outcomes are *equally likely*. Then

$$P(A) = \frac{|A|}{|S|}$$

defines a valid probability for any event  $A \subseteq S$ .

### Example (Classical probability).

- Roll a fair die.  $S = \{1, 2, 3, 4, 5, 6\}$ ,  $|S| = 6$ . For the event “even”,

$$P(\text{even}) = \frac{|\{2, 4, 6\}|}{|S|} = \frac{3}{6} = \frac{1}{2}.$$

- Deal 4 cards from a 52-card deck (all 4-card hands equally likely). Let  $A$  be the event “exactly 2 queens”.

$$P(A) = \frac{\binom{4}{2} \binom{48}{2}}{\binom{52}{4}} \approx 0.025.$$

## Example: a loaded die

- Same sample space  $S = \{1, 2, 3, 4, 5, 6\}$  as for the fair die, but a different probability measure:

$$P(\{\omega\}) = \begin{cases} 1/5 & \text{if } \omega \in \{1, 2, 3, 4, 6\}, \\ 0 & \text{if } \omega = 5. \end{cases}$$

- Compare with the fair die (where each outcome had probability  $1/6$ ).
- Now, for example,

$$P(\text{even}) = P(\{2, 4, 6\}) = \frac{3}{5}, \quad P(\text{not a 4}) = 1 - P(\{4\}) = \frac{4}{5}.$$

- Same sample space, different probability measure  $\Rightarrow$  different model of uncertainty.

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# Definition

Conditional probability is about updating our predictions with new information.

As before, let  $\Omega$  denote the outcome space and  $E$  the event space.

## Definition 3 (Conditional probability).

Let  $A, B \in E$  be two events. If  $\mathbb{P}(B) > 0$  then the **conditional probability of  $A$  given  $B$**  is

$$\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Important note: in general  $\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A)$ .

## Conditional probability from a $2 \times 2$ table

Suppose in a sample of size  $N$  we record whether two events  $A$  and  $B$  occur.

|       | $A$ | $A^c$ |
|-------|-----|-------|
| $B$   | $a$ | $b$   |
| $B^c$ | $c$ | $d$   |

Total sample size:  $N = a + b + c + d$ .

- $P(A) = \frac{a + c}{N}$  (all cases with  $A$  divided by all cases).
- $P(B) = \frac{a + b}{N}$  (all cases with  $B$  divided by all cases).
- $P(A \cap B) = \frac{a}{N}$  (cases with both  $A$  and  $B$ ).
- $P(A | B) = \frac{\text{cases with } A \text{ and } B}{\text{all cases with } B} = \frac{a}{a + b}$ .
- $P(B | A) = \frac{\text{cases with } A \text{ and } B}{\text{all cases with } A} = \frac{a}{a + c}$ .

## Example: Donuts

Take a donut at random.



$$\mathbb{P}(\text{blue donut} | \text{donut has sprinkles}) = \frac{\mathbb{P}(\text{blue donut with sprinkles})}{\mathbb{P}(\text{donut with sprinkles})} = \frac{1}{15} / \frac{11}{15} = \frac{1}{11}$$

$$\mathbb{P}(\text{donut has sprinkles} | \text{blue donut}) = \frac{\mathbb{P}(\text{blue donut with sprinkles})}{\mathbb{P}(\text{blue donut})} = \frac{1}{15} / \frac{2}{15} = \frac{1}{2}$$

## Example: Stop and search in the UK

A police officer in the UK can stop and search you if they have 'reasonable grounds' to suspect you're carrying drugs, weapons, or stolen property.

|                             | Stop and search (self-identified ethnicity, number) |               |               |               |                  | Total (known)  | Not stated / unknown | Total (all)    |
|-----------------------------|-----------------------------------------------------|---------------|---------------|---------------|------------------|----------------|----------------------|----------------|
|                             | White                                               | Black         | Asian         | Mixed         | Chinese or Other |                |                      |                |
| No further action           | 118,138                                             | 37,233        | 20,101        | 6,859         | 3,022            | 185,353        | 22,648               | 208,001        |
| Arrest                      | 26,271                                              | 12,002        | 4,635         | 2,094         | 806              | 45,808         | 4,254                | 50,062         |
| Khat / Cannabis Warning     | 12,176                                              | 3,617         | 2,486         | 906           | 394              | 19,579         | 1,071                | 20,650         |
| Summons                     | 2,614                                               | 518           | 353           | 134           | 60               | 3,679          | 251                  | 3,930          |
| Community Resolution        | 2,900                                               | 476           | 362           | 152           | 49               | 3,939          | 327                  | 4,266          |
| Penalty Notice for Disorder | 1,904                                               | 852           | 708           | 159           | 67               | 3,690          | 201                  | 3,891          |
| Caution                     | 1,117                                               | 109           | 127           | 26            | 9                | 1,388          | 87                   | 1,475          |
| <b>Overall</b>              | <b>165,120</b>                                      | <b>54,807</b> | <b>28,772</b> | <b>10,330</b> | <b>4,407</b>     | <b>263,436</b> | <b>28,839</b>        | <b>292,275</b> |

Principal outcomes following stop and search, England and Wales, 2016/17. ONS, 'Race and the criminal justice system 2016'

$$\mathbb{P}(\text{arrest}) = \frac{45,808}{263,436} = 0.174, \text{ or roughly } 17\%.$$

$$\mathbb{P}(\text{arrest}|\text{white}) = \frac{\mathbb{P}(\text{arrest and white})}{\mathbb{P}(\text{white})} = \frac{26,271}{165,120} = 0.159, \text{ or roughly } 16\%.$$

$$\mathbb{P}(\text{arrest}|\text{black}) = \frac{\mathbb{P}(\text{arrest and black})}{\mathbb{P}(\text{black})} = \frac{12,002}{54,807} = 0.219, \text{ or roughly } 22\%.$$

## Product rule

From the definition of conditional probability, for events  $A, B$  with  $\mathbb{P}(B) > 0$ ,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B) \mathbb{P}(B).$$

Similarly, if  $\mathbb{P}(A) > 0$ ,

$$\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A) \mathbb{P}(A).$$

This is the **multiplicative** (or **product**) law of probability. It lets us write joint probabilities as “prior  $\times$  conditional”.



# The law of total probability

## Theorem 2 (Law of total probability).

Let  $A, B \in \mathcal{E}$ . Then

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c).$$

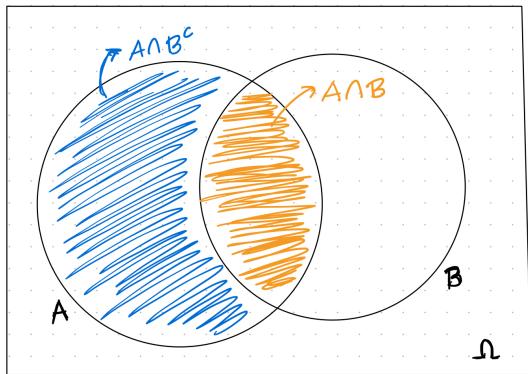
- The event  $A$  splits into two disjoint pieces:

$$A = (A \cap B) \cup (A \cap B^c).$$

- Because they are disjoint, the probabilities add:

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c).$$

- This is the simplest version of the law of total probability and is exactly what we will use in Bayes' rule.



# Bayes' Law / Rule / Theorem

## Theorem 3 (Bayes' Law).

Let  $A, B \in \mathcal{E}$  be two events with  $\mathbb{P}(A) > 0$  and  $\mathbb{P}(B) > 0$ . Then

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A) \mathbb{P}(A)}{\mathbb{P}(B)}.$$

**Idea:** Using

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \quad \text{and} \quad \mathbb{P}(A \cap B) = \mathbb{P}(B \mid A) \mathbb{P}(A),$$

we “flip” the conditioning: from  $\mathbb{P}(B \mid A)$  (likelihood) and  $\mathbb{P}(A)$  (prior) to  $\mathbb{P}(A \mid B)$  (posterior).

## Example: Gold in the Odenwald

Good news! It has been discovered that 1 in every 1000 trees in the Odenwald is made of gold. Any person that cuts down a golden tree can keep it, but anyone that cuts down a normal tree will go immediately to jail.

It's not possible to know whether a tree is gold until it has been cut down. Fortunately you have developed a test to distinguish between wooden and golden trees, and it is correct 99% of the time.

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Let's apply Bayes:

$$\mathbb{P}(\text{gold tree} \mid \text{positive test}) = \frac{\mathbb{P}(\text{gold tree}) \mathbb{P}(\text{positive test} \mid \text{gold tree})}{\mathbb{P}(\text{positive test})}$$

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$$\begin{aligned}\mathbb{P}(\text{gold tree} \mid \text{positive test}) &= \frac{\mathbb{P}(\text{gold tree}) \mathbb{P}(\text{positive test} \mid \text{gold tree})}{\mathbb{P}(\text{positive test})} \\ &= \frac{0.001 \times 0.99}{0.99 \times 0.001 + 0.01 \times 0.999} \\ &= 0.090 \dots \Rightarrow \text{So about a 9\% chance of success.}\end{aligned}$$

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# Independence

Conditional probability lets one event change the probability of another. Sometimes there is *no* effect.

## Definition 4.

Two events  $A, B \in \mathcal{E}$  are **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Equivalent formulations (when the condition makes sense):

$$\mathbb{P}(A \mid B) = \mathbb{P}(A), \quad \mathbb{P}(B \mid A) = \mathbb{P}(B).$$

**Remark.** If  $A$  and  $B$  are mutually exclusive and both have positive probability, then they *cannot* be independent.



## Example: kleine Einsteins

Albert Einstein has 3 children. Each child is equally likely to be a boy (B) or girl (G), independently of the others.

Define events:

- $A$  = “all children are the same sex”
- $B$  = “there is at most one boy”
- $C$  = “the family includes a boy and a girl”

Each of the  $2^3 = 8$  birth orders is equally likely with prob.  $1/8$ .

Quick counts:

$$\mathbb{P}(A) = \frac{2}{8}, \quad \mathbb{P}(B) = \frac{4}{8}, \quad \mathbb{P}(C) = \frac{6}{8}.$$

Intersections:

$$A \cap B = \{GGG\} \Rightarrow \mathbb{P}(A \cap B) = \frac{1}{8},$$

$$B \cap C = \{\text{families with exactly 1 boy}\} \Rightarrow \mathbb{P}(B \cap C) = \frac{3}{8}.$$

## Example ctd: kleine Einsteins

Are  $A$  and  $B$  independent?

$$\mathbb{P}(A)\mathbb{P}(B) = \frac{2}{8} \cdot \frac{4}{8} = \frac{1}{8} = \mathbb{P}(A \cap B).$$

So  $A$  and  $B$  are independent.

Are  $B$  and  $C$  independent?

$$\mathbb{P}(B)\mathbb{P}(C) = \frac{4}{8} \cdot \frac{6}{8} = \frac{3}{8} = \mathbb{P}(B \cap C).$$

So  $B$  and  $C$  are also independent.

What about  $A$  and  $C$ ?

$$A \cap C = \emptyset \Rightarrow \mathbb{P}(A \cap C) = 0, \quad \mathbb{P}(A)\mathbb{P}(C) = \frac{2}{8} \cdot \frac{6}{8} > 0.$$

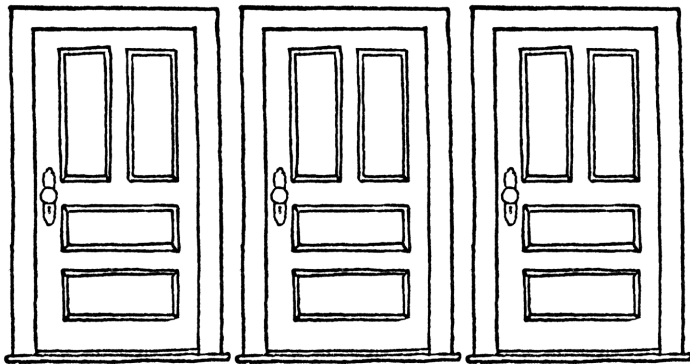
So  $A$  and  $C$  are *not* independent.

This shows: even if  $A$  is independent of  $B$  and  $B$  of  $C$ , it need not be true that  $A$  is independent of  $C$ .

# Appendix

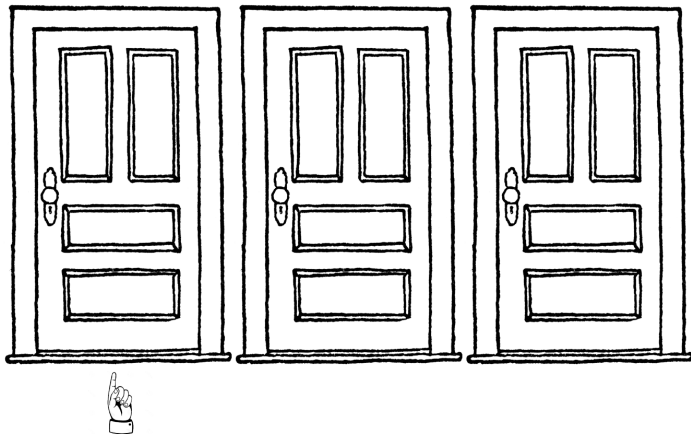
## Example: The Monty Hall Problem

You are a contestant on a game show, and have a choice of three doors. Behind one is a new car, and behind the other two are goats. You pick a door, and the host, who knows what's behind the doors, opens another door to reveal a goat. He then says **“Do you want to switch your choice of door?”** What do you do?



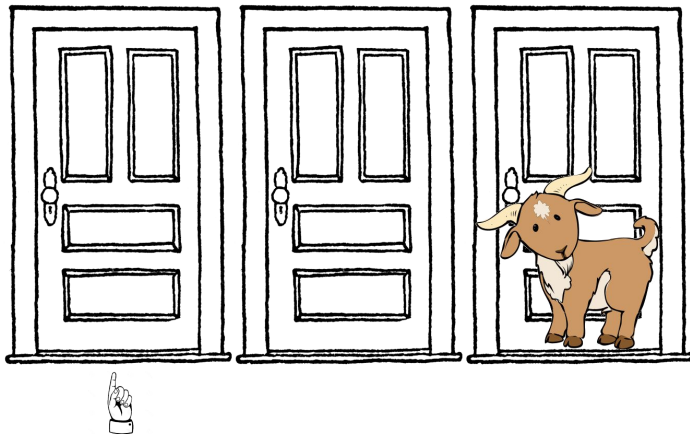
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## Example: The Monty Hall Problem

**The wrong approach:**

Define the event  $A_n = \{\text{door } n \text{ hides a car}\}$ . Then

$$\mathbb{P}(A_1|A_3^c) = \frac{\mathbb{P}(A_1 \cap A_3^c)}{\mathbb{P}(A_3^c)} = \frac{1}{3} / \frac{2}{3} = \frac{1}{2}$$

So either door has a 50% chance of hiding the car. No point in swapping!

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So either door has a 50% chance of hiding the car. No point in swapping!

*Why is this incorrect?*

Host opens door 3  $\Rightarrow$  Door 3 hides a goat

but

Host opens door 3  $\Leftarrow$  Door 3 hides a goat

We haven't used all the information available to us.



## Example: The Monty Hall Problem

### The right approach:

Remember: we chose door 1. Define the events  $A_n = \{\text{door } n \text{ hides a car}\}$  and  $B_n = \{\text{the host reveals a goat behind door } n\}$ .

Then

$$\begin{aligned}\mathbb{P}(B_3) &= \mathbb{P}(B_3 \cap A_1) + \mathbb{P}(B_3 \cap A_2) + \mathbb{P}(B_3 \cap A_3) \\ &= \mathbb{P}(B_3|A_1)\mathbb{P}(A_1) + \mathbb{P}(B_3|A_2)\mathbb{P}(A_2) + \mathbb{P}(B_3|A_3)\mathbb{P}(A_3) \\ &= \frac{1}{2} \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} = \frac{1}{2}\end{aligned}$$

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$$\mathbb{P}(A_1|B_3) = \frac{\mathbb{P}(A_1 \cap B_3)}{\mathbb{P}(B_3)} = \frac{1}{6} / \frac{1}{2} = \frac{1}{3}$$

## Simpson's paradox

Sometimes conditioning on an event gives the *opposite* results to what you would expect.

|        | Women (200) |           | Men (200) |           |
|--------|-------------|-----------|-----------|-----------|
|        | Degree      | No degree | Degree    | No degree |
| Whigs  | 90          | 35        | 10        | 45        |
| Tories | 60          | 15        | 40        | 105       |
| % Tory | 40%         | 30%       | 80%       | 70%       |

**Expectation:** Having a degree increases support for the Tories.

## Simpson's paradox

Sometimes conditioning on an event gives the *opposite* results to what you would expect.

|        | Women (200) |           | Men (200) |           | Total (400) |           |
|--------|-------------|-----------|-----------|-----------|-------------|-----------|
|        | Degree      | No degree | Degree    | No degree | Degree      | No degree |
| Whigs  | 90          | 35        | 10        | 45        | 100         | 80        |
| Tories | 60          | 15        | 40        | 105       | 100         | 120       |
| % Tory | 40%         | 30%       | 80%       | 70%       | 50%         | 60%       |

**Expectation:** Having a degree increases support for the Tories.

**Paradox:** Although **this effect** is observed in both constituent groups, the total group shows **the opposite effect**.

# Simpson's Paradox

$$\begin{aligned}\mathbb{P}(\text{Tory}|\text{degree}) &= \frac{\mathbb{P}(\text{Tory}, \text{degree})}{\mathbb{P}(\text{degree})} && \text{(def of C.P.)} \\ &= \frac{\mathbb{P}(\text{Tory}, \text{degree}, \text{woman}) + \mathbb{P}(\text{Tory}, \text{degree}, \text{man})}{\mathbb{P}(\text{degree})} && \text{(L.T.P.)} \\ &= \frac{\mathbb{P}(\text{Tory}|\text{degree}, \text{woman}) \cdot \mathbb{P}(\text{degree}, \text{woman})}{\mathbb{P}(\text{degree})} + \text{---man---} && \text{(def of C.P.)} \\ &= \mathbb{P}(\text{Tory}|\text{degree}, \text{woman})\mathbb{P}(\text{woman}|\text{degree}) + \mathbb{P}(\text{Tory}|\text{degree}, \text{man})\mathbb{P}(\text{man}|\text{degree}) && \text{(def of C.P.)}\end{aligned}$$

Similarly

$$\begin{aligned}\mathbb{P}(\text{Tory}|\text{no degree}) &= \mathbb{P}(\text{Tory}|\text{no degree}, \text{woman})\mathbb{P}(\text{woman}|\text{no degree}) \\ &\quad + \mathbb{P}(\text{Tory}|\text{no degree}, \text{man})\mathbb{P}(\text{man}|\text{no degree})\end{aligned}$$

# Simpson's Paradox

**Conclusion:** The unequal weighting of men and women with degrees hides (and even seems to reverse!) the effect that a degree has on voting preferences.

**Important:** Be aware of dependencies within data. Correlations that seem irrelevant can alter results significantly.